As the denominator of a fraction cannot be 0, $a+kd \neq 0$ for any integer $k \geq 0$. So the condition $\frac{-a}{d}$ is not a whole number is essential. Problems in harmonic progression can be converted into problems in arithmetic progression and be solved using arithmetic progression techniques and formulae.



(i) The sequence $\left(\frac{1}{n}\right) = 1, \frac{1}{2}, \frac{1}{3}, \cdots$ is a harmonic sequence. One can draw the graph of
$\{(n, \frac{1}{n}) : n \in \mathbb{N}\}$ and visualize the harmonic sequence $\left(\frac{1}{n}\right)$.

(ii) If a, b, c are in HP, then $b = \dfrac{2ac}{a+c}$.

(iii) In a triangle, if the altitudes are in AP, then the sides are in HP

Example 5.12 Prove that if a, b, c are in HP, if and only if $\frac{a}{c} = \frac{a-b}{b-c}$.

Solution:
If a, b, c are in HP, then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP. Thus, we have $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$, which gives $ab - ac = ac - bc$. So, $a(b-c) = c(a-b)$, which gives $\frac{a}{c} = \frac{a-b}{b-c}$. On the other hand, if $\frac{a}{c} = \frac{a-b}{b-c}$, then $a(b-c) = c(a-b)$. Dividing each terms of both sides by abc, we get $\frac{1}{c} - \frac{1}{b} = \frac{1}{b} - \frac{1}{a}$. Thus, $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP and hence a, b, c are in HP.

Example 5.13 If the 5th and 9th terms of a harmonic progression are $\frac{1}{19}$ and $\frac{1}{35}$, find the 12th term of the sequence.

Solution:
Let h_n be the harmonic progression and let $a_n = \dfrac{1}{h_n}$. Then $a_5 = 19$ and $a_9 = 35$. As a_n's form an arithmetic progression, we have $a + 4d = 19$ and $a + 8d = 35$. Solving these two equations, we get $a = 3$ and $d = 4$. Thus $a_{12} = a + 11d = 47$. Thus the 12th term of the harmonic progression is $\dfrac{1}{47}$.

What can we say about constant sequences? All constant sequences other than the zero sequence are harmonic progressions also.

5.4.4 Arithmetic, Geometric and Harmonic Mean

We know the concept of "*average*". There are many "*averages*". Arithmetic mean (AM), Geometric mean(GM) and Harmonic mean(HM) are also some averages. Let us now recall the definitions of arithmetic mean and geometric mean, where the terms need not in AP or GP.

Arithmetic Mean and Geometric Mean

Definition 5.3

Let n be any positive integer. Let $a_1, a_2, a_3, \ldots, a_n$ be n numbers . Then the number

$$\frac{a_1 + a_2 + a_3 + \cdots + a_n}{n}$$

is called the ***arithmetic mean*** of the numbers $a_1, a_2, a_3, \ldots, a_n$.

The numbers $a_1, a_2, a_3, \ldots, a_n$ need not be distinct and it is not necessary that the numbers are positive. It easily follows from the definition that 16 is the arithmetic mean of the numbers $14, 14, 17, 20, 15$.